



TERM 4 CAPSTONE: FULL-STACK IOT DASHBOARD



TODAY'S
CHALLENGE: SOLVE
REAL PROBLEMS!

Water Overflow
Farm Irrigation
Air Quality Alerts

"Can we fix these
real problems with
smart dashboards?"



WHAT IS IOT? CONNECT & CONTROL!

IoT means connecting things to the internet to make life smarter!
Sensors collect data from the real world → Data travels through Wi-Fi & Cloud → Dashboards show and control everything!
From water tanks to farms to city air — IoT helps us see, understand, and act!

💡 Aarav's Tip: "Sensors collect data, dashboards show it — that's IoT magic!"





IOT LAYERS: HOW OUR SYSTEM WORKS

SENSORS / PERCEPTION LAYER

Sensors detect water levels, soil moisture, and air quality in the real world and collect data around us!



NETWORK / CLOUD GATEWAY LAYER

Data travels from sensors to the cloud using Wi-Fi.
Ananya says: "Each layer has a special job!"

APPLICATION / DASHBOARD LAYER

Dashboards display sensor data with gauges and graphs so you can see and control everything easily!



SENSORS LAYER: SENSING OUR WORLD

Sensors check water levels, soil moisture, and air quality all around us!

💧 Water sensor monitors the overhead tank in your home.

🌱 Soil moisture sensor checks if the farm field needs water.

☁️ AQI sensor measures air quality in busy cities like Delhi.

Sensors collect real data and send it to the cloud instantly!



NETWORK & CLOUD LAYER: DATA ON THE MOVE

DATA TRAVELS FROM SENSORS TO
CLOUD USING WI-FI AND GATEWAYS!

Sensors send readings wirelessly to a gateway, which passes the data to the cloud. From there, your dashboard can show it anywhere, anytime! Ananya says: "It's just like sending a message through the internet!"



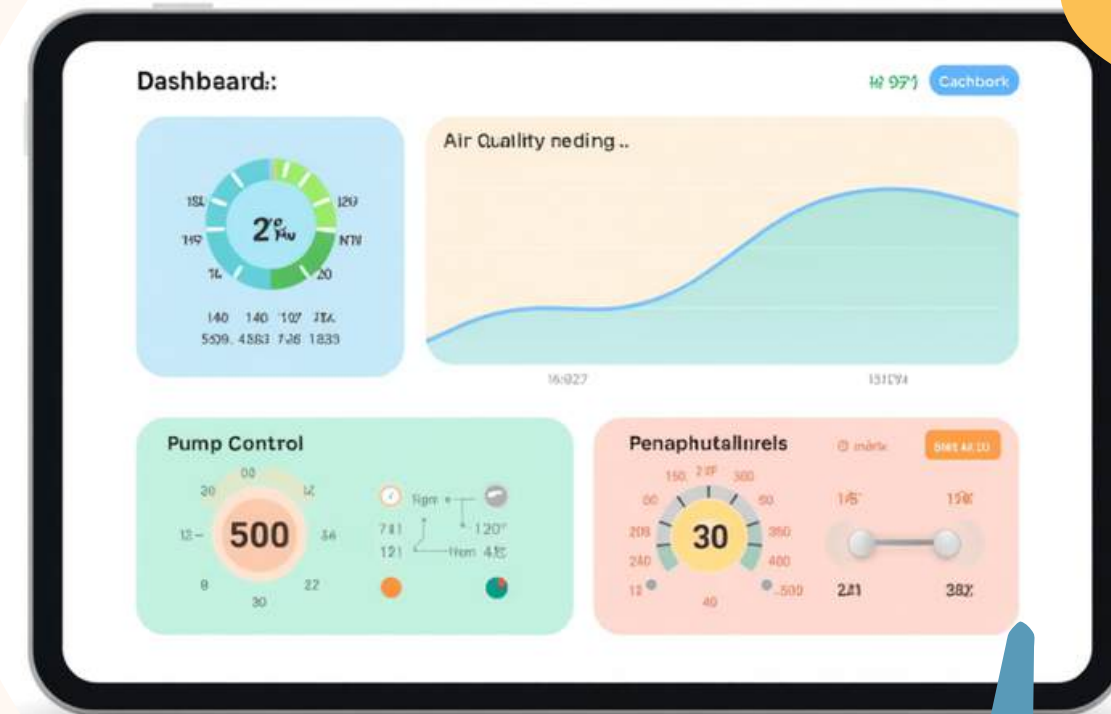
DASHBOARD LAYER

YOUR CONTROL CENTRE

Dashboards show sensor data and let you control devices. See water levels, air quality, and soil moisture — all in one place!

WIDGETS & CONTROLS

Use gauges, line graphs, status lights, and switches to monitor and control your IoT devices in real time!

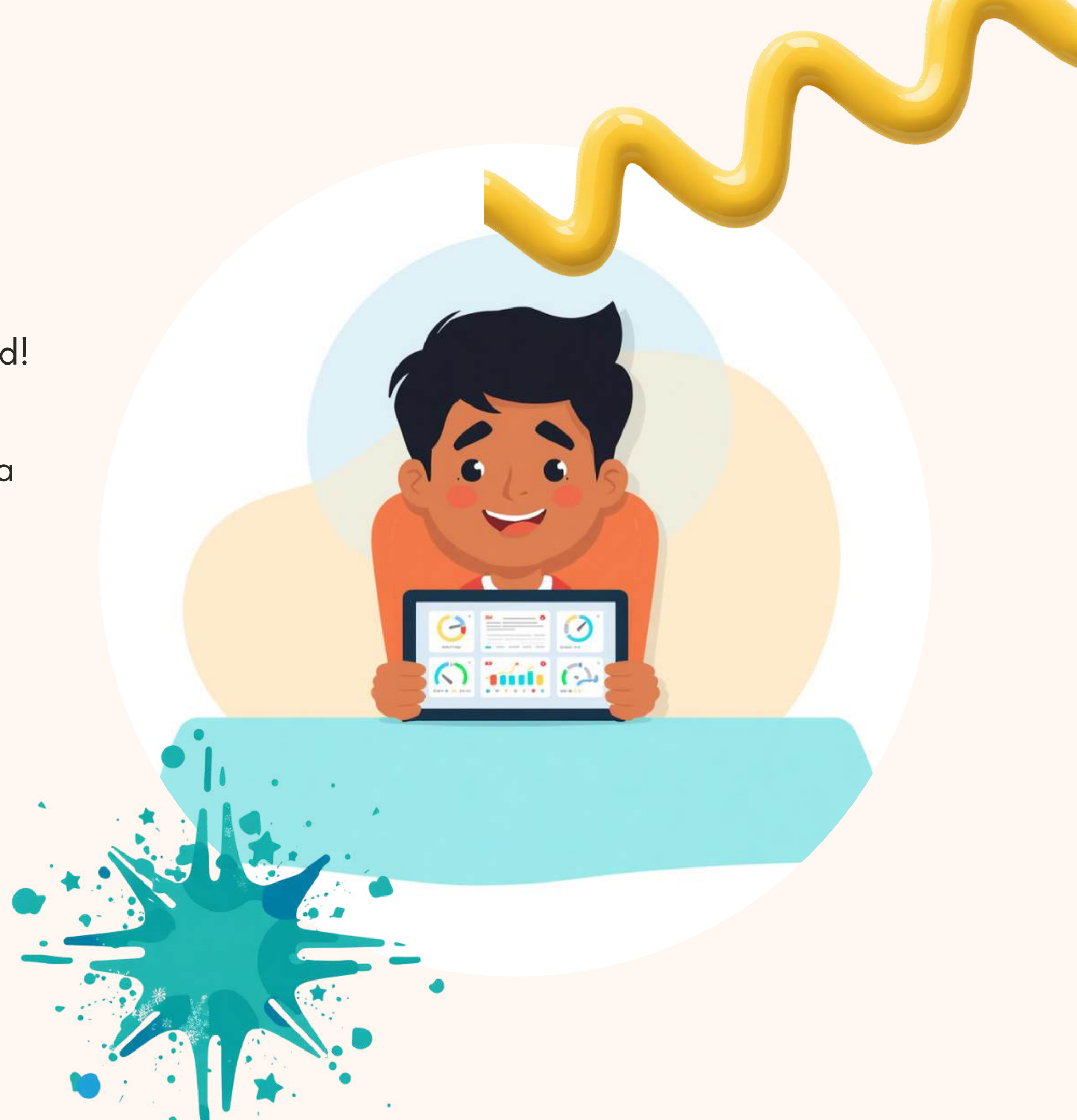


BUILD YOUR DASHBOARD:

STEP 1 — LAYOUT & UI

Pick gauges, graphs, and switches for your dashboard! Use a drag-and-drop UI builder to arrange your widgets. Choose the right widget for each sensor — a gauge for water level, a line graph for temperature, and a switch to control your pump!

👤 Aarav's Tip: "Keep it simple and clear — a good dashboard is easy to read at a glance!"



STEP 2: BIND SENSOR DATA

CONNECT SENSORS TO WIDGETS

Link your sensor readings to dashboard widgets like gauges and graphs. When the sensor sends data, the widget updates automatically — like magic!



ANANYA'S TIP

"Watch your data come alive! Each widget is connected to a real sensor — water level, soil moisture, or air quality. Change the sensor, see the dashboard update!"

— Ananya

BUILD YOUR DASHBOARD: STEP 3 — STATUS INDICATORS

ADD LIGHTS TO SHOW SAFE OR WARNING LEVELS!

Add green and red status lights to your dashboard. A green light means everything is safe — water levels are normal and air quality is good. A red light means WARNING — the tank is overflowing or Delhi AQI is too high! Indicators help you act fast before problems get worse.

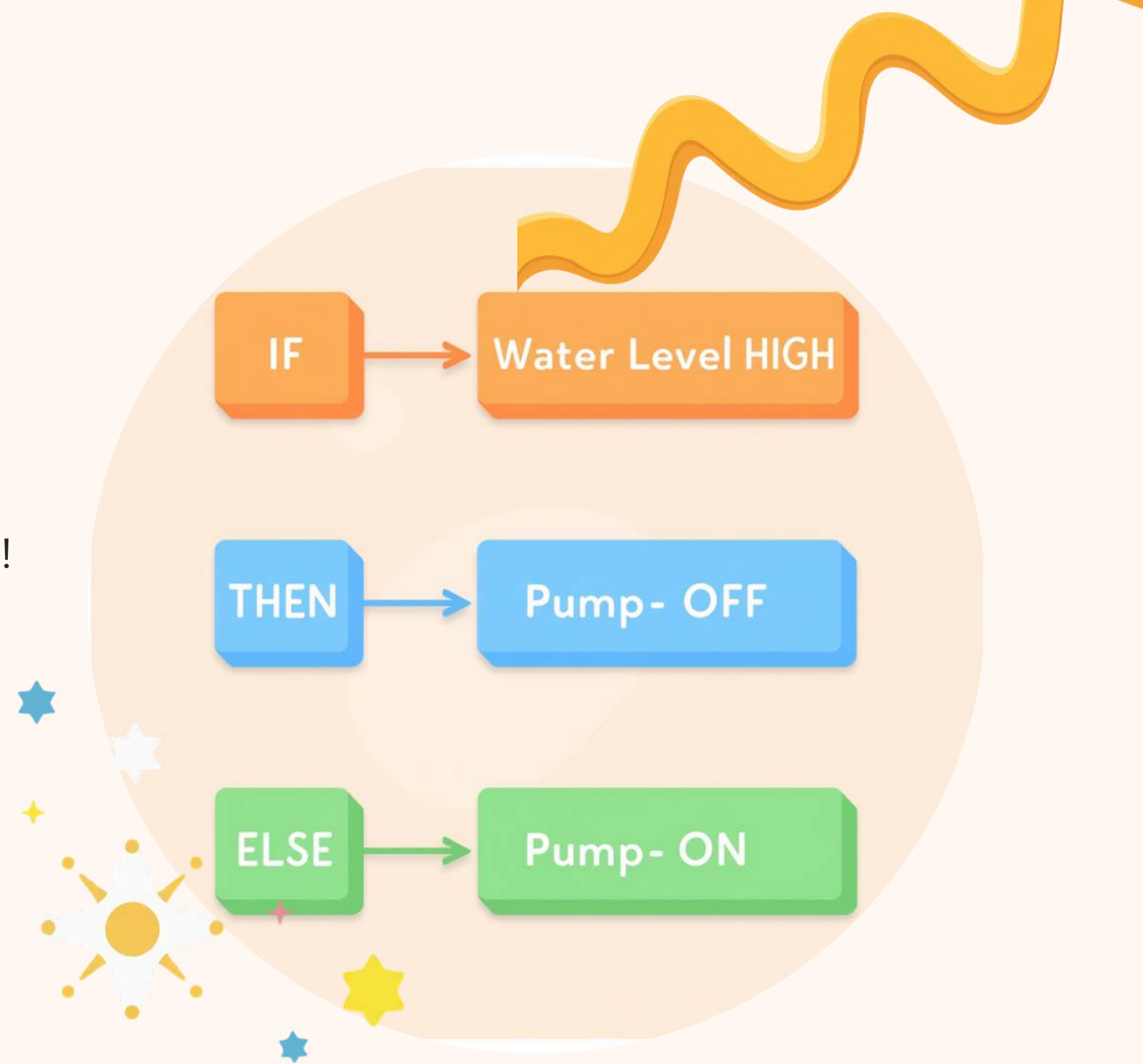


AUTOMATION LOGIC!

Use IF-THEN-ELSE blocks to automate pumps and alerts!

When sensors detect high water levels, your code can automatically turn the pump OFF – no human needed! Block coding makes your IoT system truly smart.

Aarav's tip: "If water's high, then turn pump OFF!"



TEST YOUR SYSTEM!

RUN YOUR TESTS

Test your sensors, dashboard, and automation blocks all together. Check if your water level gauge updates, your AQI alerts trigger, and your pump automation works correctly!

FIND AND FIX BUGS!

If something doesn't work, that's okay — debugging is part of building! Check your sensor connections, widget bindings, and IF-THEN-ELSE blocks one step at a time.



PRESENT YOUR DASHBOARD!

SHARE YOUR SMART SOLUTION
WITH THE WORLD!

Show your dashboard and explain how it helps your community. Walk through each part — sensors, data, and automation. Ananya's tip: "Tell your story clearly!" ✨





RECAP & KEEP INNOVATING!



IOT CONNECTS EVERYTHING!

Sensors collect data and send it to the cloud. Dashboards show it all in one place. IoT solves real Indian problems!

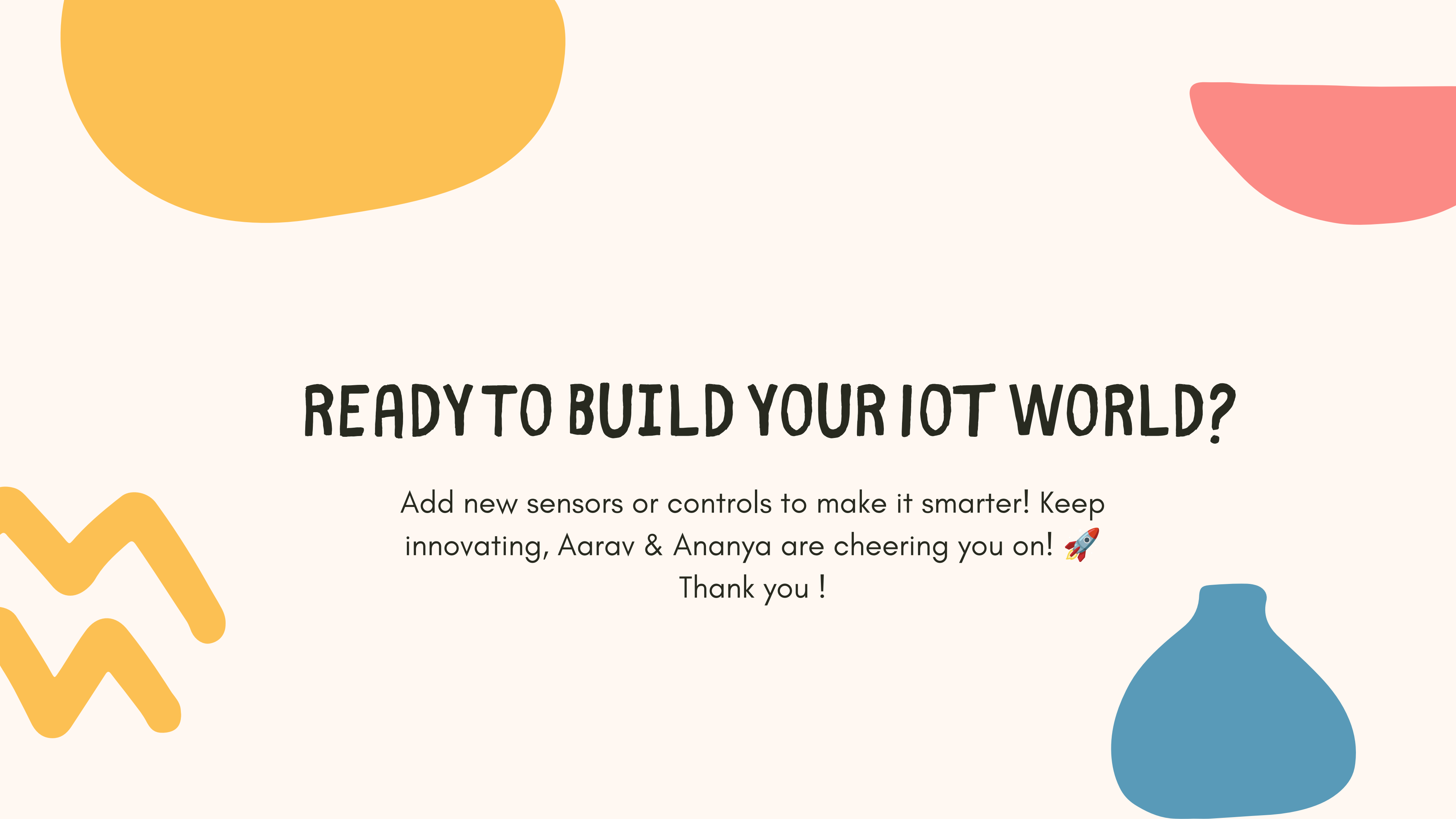
SMART DASHBOARDS SAVE THE DAY!

Build clear dashboards with gauges, graphs, and status lights. Smart alerts warn you before overflow or poor air quality!

AUTOMATION MAKES LIFE EASIER!

IF-THEN-ELSE blocks automate pumps and alerts. Systems work on their own to keep water, farms, and air safe!





READY TO BUILD YOUR IOT WORLD?

Add new sensors or controls to make it smarter! Keep innovating, Aarav & Ananya are cheering you on! 🚀

Thank you !